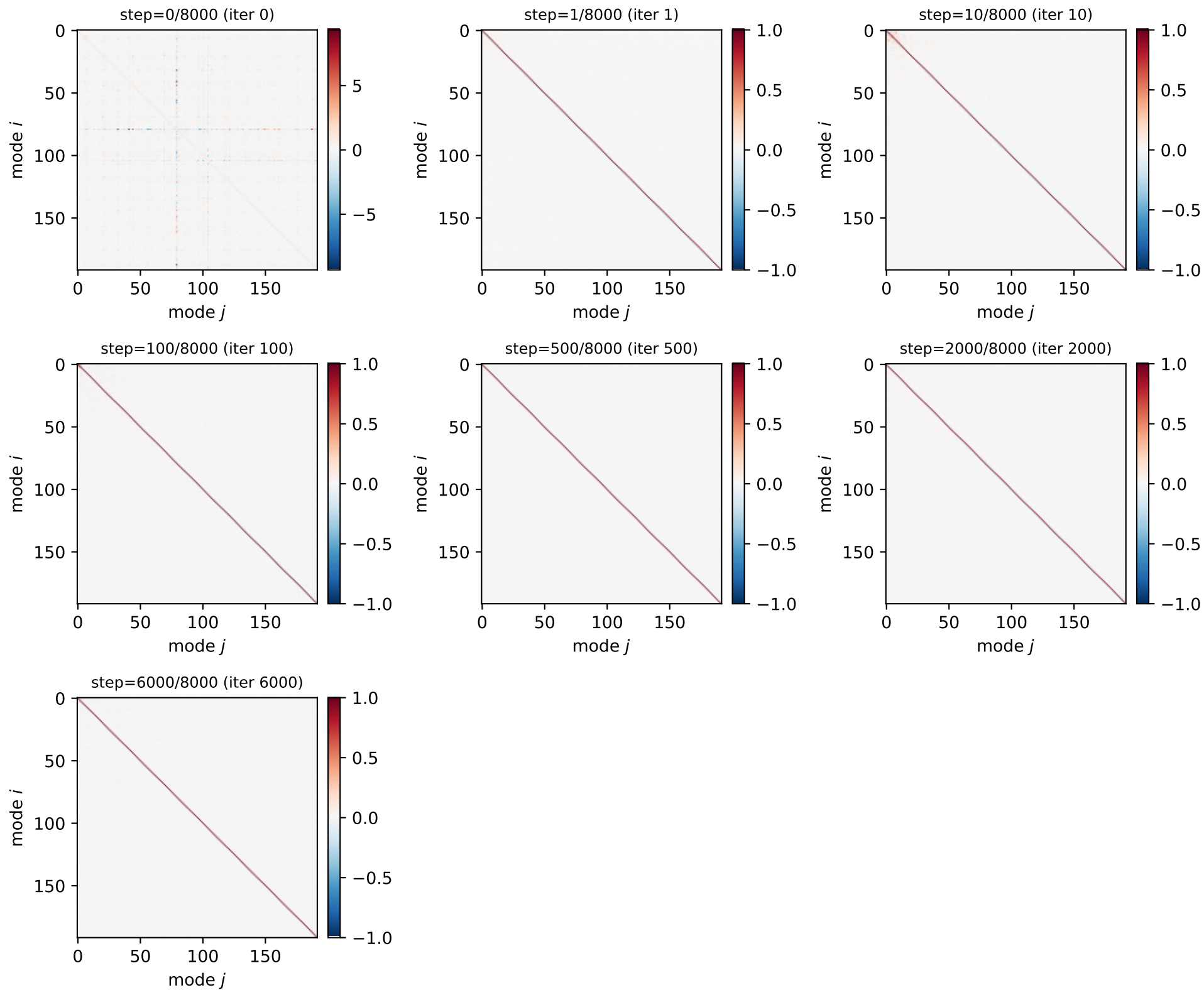
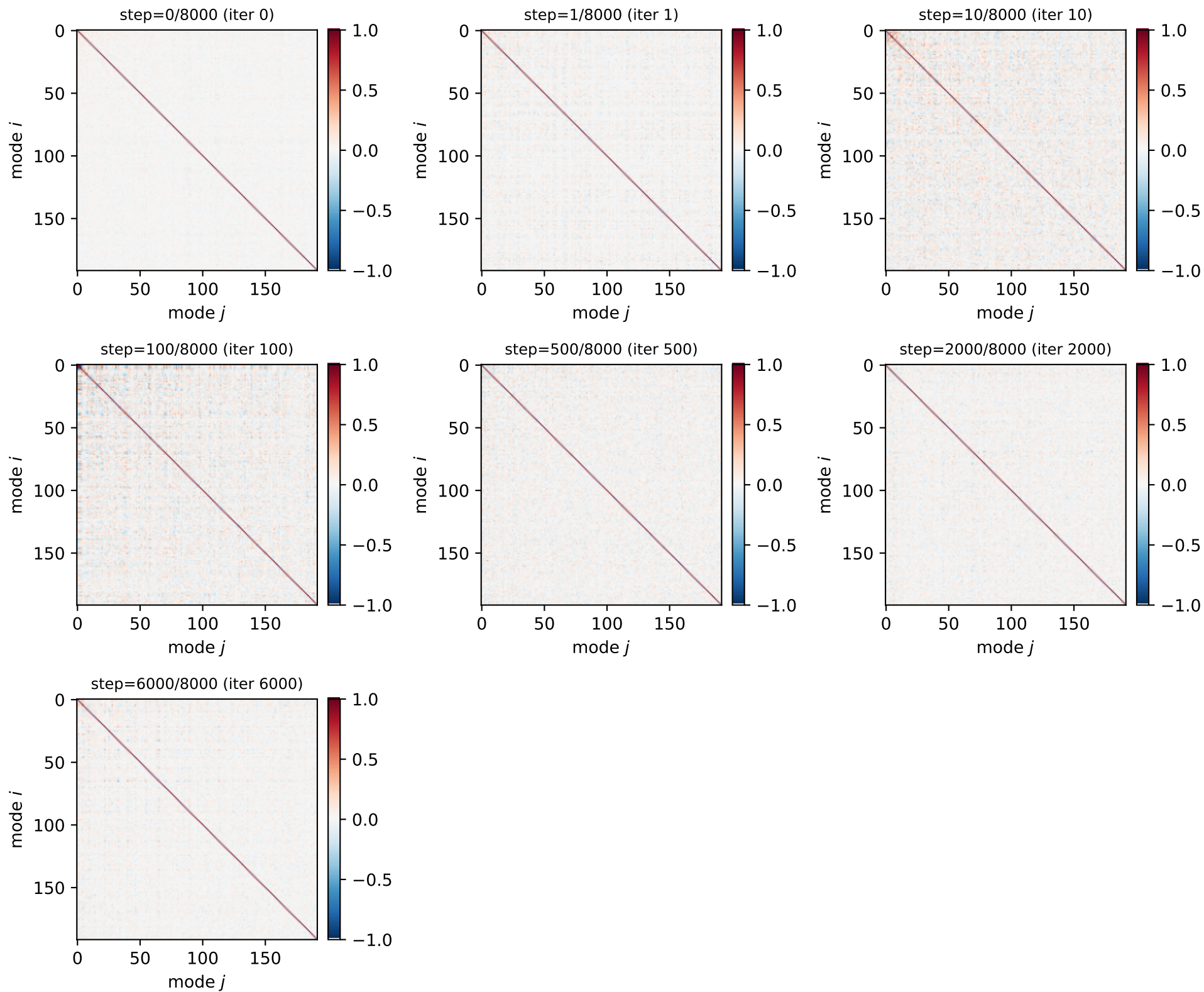


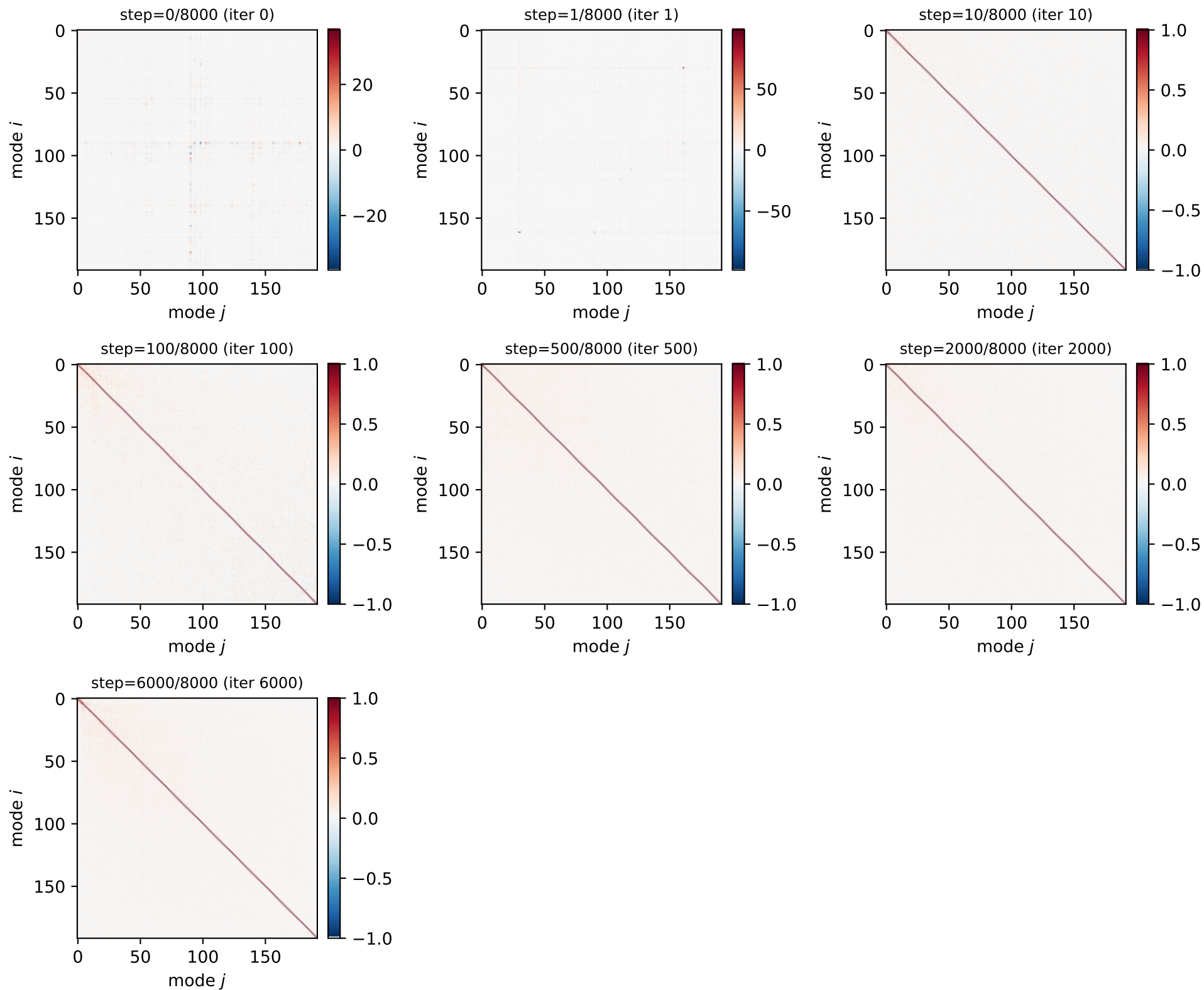
Attention value projection:  $\gamma_{ij}/\sqrt{|\gamma_{ii}\gamma_{jj}|}$  at each group's own fork step



Attention value projection: residual cross-correlation  $\eta_{ij}^\perp = \langle Q_i, Q_j \rangle$  at each group's own fork step



MLP up-projection:  $\gamma_{ij}/\sqrt{|\gamma_{ii}\gamma_{jj}|}$  at each group's own fork step





MLP up-projection: residual cross-correlation  $\eta_{ij}^\perp = \langle Q_i, Q_j \rangle$  at each group's own fork step

